

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AMA577
Subject Title	Theoretical Aspects of Reinforcement Learning
Credit Value	3
Level	5
Pre-requisite / Co-requisite/ Exclusion	Pre-requisite: AMA564 Deep Learning
Objectives	The objective of this course is to equip students with the mathematical foundations and practical skills of Reinforcement Learning (RL), a core subfield of artificial intelligence. The curriculum will cover mathematical foundations of RL, key RL algorithms, computational tools, and real-world applications, emphasizing both the theoretical foundations and problem-solving skills in dynamic environments. Students will learn to analyze the benefits, challenges, and ethical considerations of RL deployment in various industries.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) Know the foundation concepts of reinforcement learning and the relevant algorithms b) Understand various RL algorithms (e.g., Q-learning, Policy Gradients, DQN) and their strengths and weaknesses. c) Apply the reinforcement learning algorithms for practical problems using programming frameworks (e.g., PyTorch, TensorFlow). d) Derive insights from RL real-world applications in domains such as robotics, gaming, healthcare, and autonomous systems.
Subject Synopsis/ Indicative Syllabus	Key topics to be delivered: a) The development history of RL and its integration with deep learning. b) Mathematical foundations and core RL concepts: state spaces, action spaces, reward functions, Markov Decision Processes (MDPs). c) Key RL algorithms and their mathematical guarantees: Value-based methods (Q-learning, SARSA), Policy-based methods (REINFORCE, Actor-Critic), Model-based RL. d) Advanced RL techniques: exploration-exploitation trade-offs, experience replay, multi-agent RL, hierarchical RL, RLHF e) Case studies: RL in game AI (e.g., AlphaGo), robotic control, recommendation systems, and resource optimization. f) Selected advanced topics: sample efficiency, convergence stability, reward engineering, federated RL, and ethical considerations.

Teaching/Learning Methodology	<u>Lectures:</u> Introduce RL theory, algorithm derivations, and industry applications, with a focus on mathematical foundations and technical frameworks. <u>Tutorials:</u> Hands-on coding sessions and group discussions on case studies, using real-world datasets and simulation environments (e.g., OpenAI Gym, MuJoCo).																																							
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="4">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>1. Project</td><td>40%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Assignments</td><td>20%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>3. Exam</td><td>40%</td><td>✓</td><td>✓</td><td></td><td>✓</td></tr><tr><td>Total</td><td>100%</td><td colspan="4"></td></tr></table>						Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				a	b	c	d	1. Project	40%	✓	✓	✓	✓	2. Assignments	20%	✓	✓	✓	✓	3. Exam	40%	✓	✓		✓	Total	100%				
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	1. Project	40%	✓	✓	✓	✓																																		
	2. Assignments	20%	✓	✓	✓	✓																																		
	3. Exam	40%	✓	✓		✓																																		
	Total	100%																																						
Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:																																								
<u>Project:</u> Students will design and implement an RL agent for a specific domain, demonstrating their ability to apply algorithms, analyze results, and address real-world challenges.																																								
<u>Assignments</u> Assignments covering RL foundational concepts and algorithms and scenario-based applications, enhancing students’ understanding of core concepts and algorithms and application of RL.																																								
<u>Exam</u> Written assessment covering RL theory, algorithm comparisons, and scenario-based problem-solving, evaluating students’ understanding of core concepts and their application to practical scenarios.																																								
Student Study Effort Expected	Class contact:																																							
	▪ Lecture			26 Hrs.																																				
	▪ Tutorial			13 Hrs.																																				
	Other student study effort:																																							
	▪ Self-learning from reference materials			40 Hrs.																																				
	▪ Reading and research for the case study			40 Hrs.																																				
	Total student study effort			119 Hrs.																																				

Reading List and References	[1] Sutton, R. S., & Barto, A. G. (2020). <i>Reinforcement Learning: An Introduction</i> (2nd ed.). MIT Press. [2] Li, L., & Chen, Y. (2023). <i>Deep Reinforcement Learning: Principles, Algorithms, and Applications</i>. Springer. [3] Arulkumaran, K., Deisenroth, M. P., Brundage, M., & Bharath, A. A. (2017). Deep reinforcement learning: A brief survey. <i>IEEE Signal Processing Magazine</i>, 34(6), 26–38. [4] Szepesvári, C. (2022). <i>Algorithms for reinforcement learning</i>. Springer nature. [5] McKinsey & Company. (2024). <i>The Future of Reinforcement Learning in Industry</i>.
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